TP01(Data Structure and Algorithms)

Link: https://github.com/Mookneii/Data_Structure_And_Algorithms.git

1.

```cpp
#include <iostream>
using namespace std;
string name, major;
int age;
char gender;
int main(){
    cout<<"Please Enter:\n";
    cout<<"Name: ";cin>>name;
    cout<<"Gender(only: 'M' or 'F'): ";cin>> gender;
    cout<<"Major: "; cin>> major;
    cout<<"Age: "; cin>> age;
    if( age>=18 && gender == 'M'){
        cout<<"Hi, Mr."<<name<<"! your age is "<<age<<" year olds and you learn "<<major<<endl;
        cout<<"You can vote.";
    }else if(age >=18 && gender == 'F'){
        cout<<"Hi, Mrs."<<name<<"! your age is "<<age<<" year olds and you learn "<<major<<endl;
         cout<<"You can not vote.";
    }
    else{
        cout<<"Invalid Input!";
    }

    return 0;
}
```

```
Please Enter:
Name: Mony
Gender(only: 'M' or 'F'): M
Major: math
Age: 19
Hi, Mr.Mony! your age is 19 year olds and you learn math
You can vote.
PS D:\Data_Structure_and_Programming\Lab1>
```

2.

```cpp
#include <iostream>
using namespace std;
int main()
{
   char ch;
   cout<<"Enter one character: ";cin>>ch;
    int ascii = ch;
    if(ascii == 97 || ascii == 101 || ascii == 105 || ascii == 111 || ascii == 117 ||
           ascii == 65 || ascii == 69 || ascii == 73 || ascii == 79 || ascii == 85){
            cout<<ch<<" is a vowel."<<endl;

           }
           else{
            cout<<ch<<" is a consonant.\n";
           }

    return 0;
} // namespace std;

```

```
Enter one character: E
E is a vowel.
```

```
Enter one character: x
x is a consonant.
```

3.

```cpp
#include <iostream>
using namespace std;
int main()
{
    string name;
    char gender;
    double salary;
    cout<<"Tax system: \n";
    cout<<"Please Enter Your Name: ";cin>> name;
    cout<<"Please Enter Your gender(M / F): ";cin>>gender;
    cout<<"Please Enter Your salary: ";cin>>salary;
    if( gender ='M'){
        if(salary > 1000){
            cout<<"Your salary after taxed is: "<< salary + (salary* (9.5/100))<<"$"<<endl;
        }else if(salary >500 && salary <=1000){
            cout<<"Your salary after taxed is: "<< salary + (salary* (7/100))<<"$"<<endl;
        }else if(salary >=300 && salary <=500){
            cout<<"Your salary after taxed is: "<< salary + (salary* (5/100))<<"$"<<endl;
        }else{
            cout<<"Your salary after taxed is: "<< salary<<"$"<<"(no need to pay tax)"<<endl;
        }
    }else if(gender == 'F'){
        if(salary > 1000){
            cout<<"Your salary after taxed is: "<< salary + (salary* (8/100))<<"$"<<endl;
        }else if(salary >500 && salary <=1000){
            cout<<"Your salary after taxed is: "<< salary + (salary* (6.5/100))<<"$"<<endl;
        }else if(salary >=300 && salary <=500){
            cout<<"Your salary after taxed is: "<< salary + (salary* (3.5/100))<<"$"<<endl;
        }else{
            cout<<"Your salary after taxed is: "<< salary<<"$"<<"(no need to pay tax)"<<endl;
        }
    }

    return 0;
}
```

```
Tax system:
Please Enter Your Name: KOko
Please Enter Your gender(M / F): F
Please Enter Your salary: 3200
Your salary after taxed is: 3504$
```

4.

```cpp
#include <iostream>
#include<cstdlib>
#include<ctime>
using namespace std;
int main()
{
  int uChoice,comChoice;
    cout << "Rock Paper Scissors Game\n";
    cout << "1. Rock\n";
    cout << "2. Paper\n";
    cout << "3. Scissors\n";
    cout << "Enter your choice: ";
    cin >> uChoice;
    srand(time(0));
    comChoice = rand() % 3+1;
     if(comChoice == 1)
        cout << "Rock\n";
    else if(comChoice == 2)
        cout << "Paper\n";
    else
        cout << "Scissors\n";
        if(uChoice == comChoice){
        cout << "It's a Draw!\n";
    }
    else if((uChoice == 1 && comChoice == 3) ||
            (uChoice == 2 && comChoice == 1) ||
            (uChoice == 3 && comChoice == 2)){
        cout << "You Win!\n";
    }
    else{
        cout << "Computer Wins!\n";
    }

    return 0;
} // namespace std;
```

```
Rock Paper Scissors Game
1. Rock
2. Paper
3. Scissors
Enter your choice: 1
Rock
It's a Draw!
```

5.

```cpp
#include <iostream>
using namespace std;
int main()
{
    int min;
    cout<<"Please enter n minute: ";cin>>min;
    int hours = min/60;
    int remaindermin = min % 60;
    int sec = 0;
    cout<<"Format time(h:m:s): "<<hours<<" : "<< remaindermin<<" : "<< sec;
    return 0;
} // namespace std;

```

```
Please enter n minute: 98
Format time(h:m:s): 1 : 38 : 0
```

6.

```cpp
#include <iostream>
using namespace std;
int main()
{
   int n, sums=0;

   cout<<"summation of numbers from 1 to n.Please enter n:\n";
   cin>>n;
   if(n > 50){
   for(int i=1; i<=n;i++){
    if(i==10 or i==30){
        continue;
    }
    sums= sums+i;
   }
   cout<<"Summation from 1 to "<<n<<" is "<< sums;
   }
   else{
    cout<<"n must be greater than 50!";
   }

    return 0;
}
```

```
summation of numbers from 1 to n.Please enter n:
65
Summation from 1 to 65 is 2105
```

7.

```cpp
#include <iostream>
#include <string>
using namespace std;

struct Book {
    int bookID;
    string bookISBN;
    string bookTitle;
    int publishedYear;
    string authorName[5];
    int authorCount;
    double price;
};

void displayBookByISBN(Book books[], int size, string isbn) {
    bool found = false;

    for (int i = 0; i < size; i++) {
        if (books[i].bookISBN == isbn) {
            cout << "==================================================\n";
            cout << "Book ID: " << books[i].bookID << endl;
            cout << "Book ISBN: " << books[i].bookISBN << endl;
            cout << "Book Title: " << books[i].bookTitle << endl;
            cout << "Published Year: " << books[i].publishedYear << endl;
            cout << "Author Name: ";
            for (int j = 0; j < books[i].authorCount; j++) {
                cout << books[i].authorName[j] << " ";
            }
            cout << "\nBook Price: " << books[i].price << " $" << endl;
            cout << "==================================================\n";
            found = true;
        }
    }

    if (!found) {
        cout << "Book not found.\n";
    }
}

void displayAllBooks(Book books[], int size) {
    for (int i = 0; i < size; i++) {
        cout << "==================================================\n";
        cout << "Book ID: " << books[i].bookID << endl;
        cout << "Book ISBN: " << books[i].bookISBN << endl;
        cout << "Book Title: " << books[i].bookTitle << endl;
        cout << "Published Year: " << books[i].publishedYear << endl;
        cout << "Author Name: ";
        for (int j = 0; j < books[i].authorCount; j++) {
            cout << books[i].authorName[j] << " ";
        }
        cout << "\nBook Price: " << books[i].price << " $" << endl;
        cout << "==================================================\n";
    }
}

int main() {
    Book books[2];

    for (int i = 0; i < 2; i++) {
        cout << "\nEnter information for Book " << i + 1 << endl;

        cout << "Book ID: ";
        cin >> books[i].bookID;

        cout << "Book Title: ";
        cin >> books[i].bookTitle;

        cout << "Published Year: ";
        cin >> books[i].publishedYear;

        cout << "Book ISBN: ";
        cin >> books[i].bookISBN;

        cout << "Number of authors: ";
        cin >> books[i].authorCount;

        cout << "Author names:\n";
        for (int j = 0; j < books[i].authorCount; j++) {
            cin >> books[i].authorName[j];
        }

        cout << "Book Price: ";
        cin >> books[i].price;
    }

    cout << "\nDisplay all books:\n";
    displayAllBooks(books, 2);

    string search;
    cout << "\nSearch ISBN for book: ";
    cin >> search;
    displayBookByISBN(books, 2, search);

    return 0;
}
```

```
aa3
Book Price: 23

Display all books:
==================================================
Book ID: 1
Book ISBN: oo1
Book Title: book1
Published Year: 2001
Author Name: aa1 aa2
Book Price: 32 $
==================================================
```

```
 Published Year: 2001
 Author Name: aa1 aa2
 Book Price: 32 $
 ==================================================
 ==================================================
 Book ID: 2
 Book ISBN: oo2
 Book Title: book2
 Published Year: 2002
 Author Name: aa3
 Book Price: 23 $
 ==================================================
```

```
Author Name: aa3
Book Price: 23 $
==================================================

Search ISBN for book: oo1
==================================================
Book ID: 1
Book ISBN: oo1
Book Title: book1
Published Year: 2001
Author Name: aa1 aa2
Book Price: 32 $
==================================================
```

8.

```cpp
#include <iostream>
#include <cmath>
using namespace std;
double ConvertToFahrenheit(double Celsius){
    double Fahrenheit;
    return  Fahrenheit = (Celsius * 9.0 / 5.0) + 32;
}
double ConvertToCelsius(double Fahrenheit){
    double Celsius;
     return Celsius = (Fahrenheit - 32) * 5.0 / 9.0;
}
void findRoot(double a,double b, double c){
    double  delta = b*b - 4*a*c;
    if(delta > 0){
        double x1 = (-b + sqrt(delta))/(2 * a);
         double x2 = (-b - sqrt(delta))/(2 * a);
        cout<<"The real root X1= "<<x1<<" X2= "<<x2<<endl;
    }else if(delta == 0){
        double x = -b/(2 * a);
        cout<<"The root is x = "<<x<<endl;
    }else{
        cout<<"No Roots!"<<endl;
    }
}
void bmiCalculate( double weight, double height){
    double bmi = weight / (height * height);
    if(bmi< 18.5){
        cout<<"Underweight\n";
    }else if(bmi< 25){
        cout<<"Normal weight\n";
    }else if(bmi<30){
        cout<<"Overweight\n";
    }else{
        cout<<"Obese";
    }
}
int sumExceptDiv3(int n) {
    int sum = 0;
    for (int i = 1; i <= n; i++) {
        if (i % 3 == 0)
            continue;
        sum += i;
    }
    return sum;
}
int main(){
    int choice;

    do {
        cout << "\n===== MENU =====\n";
        cout << "1. Celsius to Fahrenheit\n";
        cout << "2. Fahrenheit to Celsius\n";
        cout << "3. Root of quadratic equation\n";
        cout << "4. Compute BMI\n";
        cout << "5. Sum from 1 to n except numbers divisible by 3\n";
        cout << "0. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1: {
                double celsius;
                cout << "Enter Celsius: ";
                cin >> celsius;
                cout << "Fahrenheit = " << ConvertToFahrenheit(celsius) << endl;
                break;
            }

            case 2: {
                double fahrenheit;
                cout << "Enter Fahrenheit: ";
                cin >> fahrenheit;
                cout << "Celsius = " << ConvertToCelsius(fahrenheit) << endl;
                break;
            }

            case 3: {
                double a, b, c;
                cout << "Enter a, b, c: ";
                cin >> a >> b >> c;
                findRoot(a, b, c);
                break;
            }

            case 4: {
                double weight, height;
                cout << "Enter weight (kg): ";
                cin >> weight;
                cout << "Enter height (m): ";
                cin >> height;
                bmiCalculate(weight, height);
                break;
            }

            case 5: {
                int n;
                cout << "Enter n: ";
                cin >> n;
                cout << "Sum = " << sumExceptDiv3(n) << endl;
                break;
            }

            case 0:
                cout << "Program ended.\n";
                break;

            default:
                cout << "Invalid choice. Please try again.\n";
        }

    } while (choice != 0);

    return 0;
}
```

```
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Root of quadratic equation
4. Compute BMI
5. Sum from 1 to n except numbers divisible by 3
0. Exit
Enter your choice: 1
Enter Celsius: 100
Fahrenheit = 212
```

```
Enter your choice: 2
Enter Fahrenheit: 100
Celsius = 37.7778
```

```
Enter your choice: 3
Enter a, b, c: 1 -3 2
The real root X1= 2 X2= 1
```

```
Enter your choice: 4
Enter weight (kg): 67
Enter height (m): 1.8
Normal weight
```

```
Enter your choice: 5
Enter n: 123
Sum = 5043
```